

SUSTAINABLE AGRICULTURE

FOOD SECURITY

Smart Solar Cold Storage System

IoT-Enabled Post-Harvest Preservation for Rural Farmers

Problem Statement ID

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Problem Statement Title

Post-harvest losses in rural agriculture

Theme

Sustainable Agriculture & Food Security

PS Category

Agricultural Technology

Team ID

[leave space for user to fill]

Team Name

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Idea & Proposed Solution

IDEA TITLE

Smart Solar Cold Storage System

PROPOSED SOLUTION

An IoT-enabled, solar-powered cold storage unit designed for small-scale farmers to preserve produce quality and reduce post-harvest losses

DETAILED EXPLANATION

The system combines passive insulation with active Peltier cooling, powered entirely by solar energy with battery backup. Real-time monitoring via ESP32 tracks temperature, humidity, and door status, enabling automatic cooling control and remote alerts.

HOW IT ADDRESSES THE PROBLEM

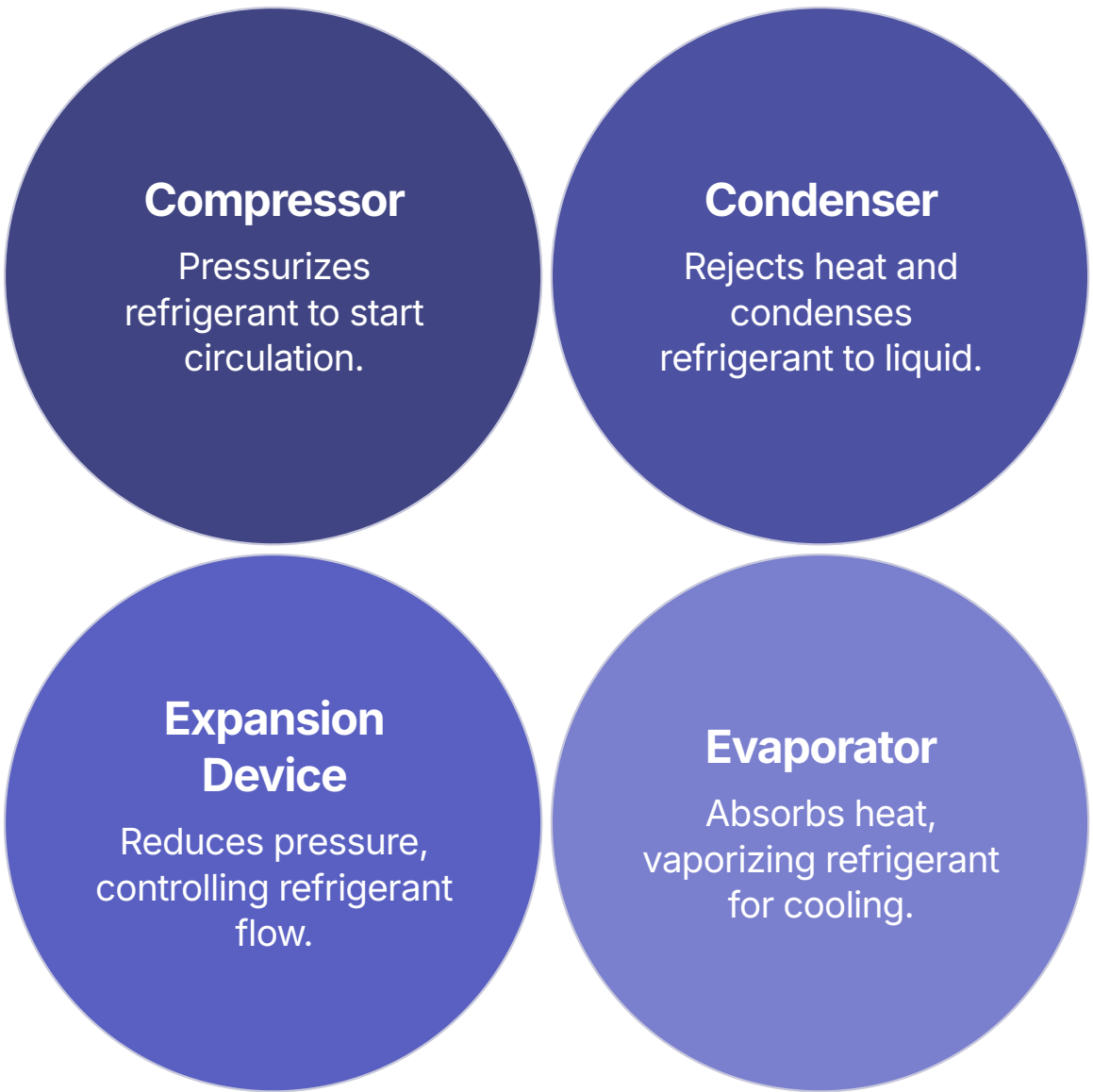
Eliminates dependency on distant cold chain facilities, reduces spoilage by 40-60%, enables farmers to store produce longer and sell at better market prices

INNOVATION & UNIQUENESS

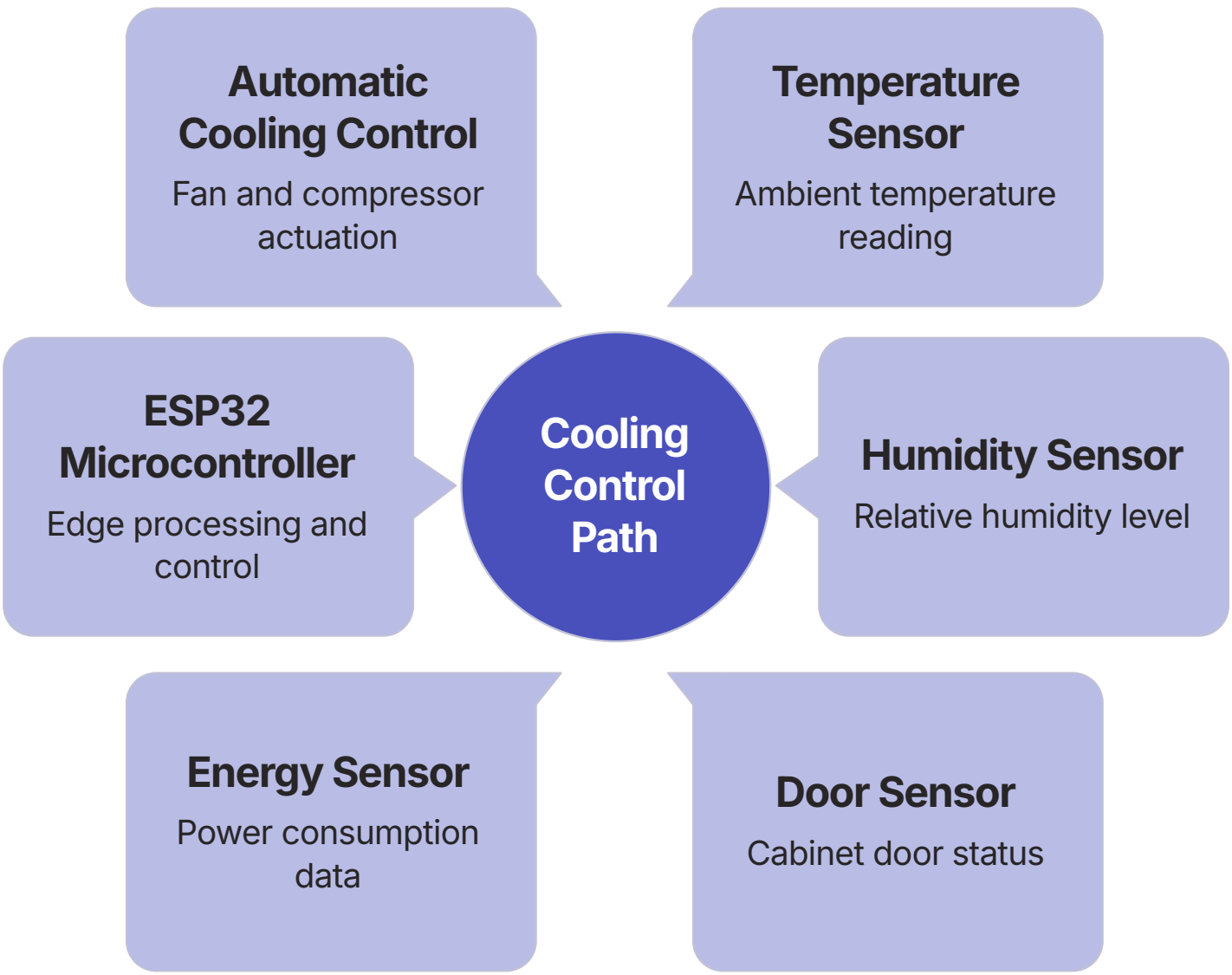
First solar-powered, IoT-enabled cold storage designed specifically for 20-40L farm-level use; combines affordability (₹25,000) with smart monitoring; DC-powered architecture for maximum efficiency

Technical Approach

❄️ COOLING PATH



🔌 CONTROL PATH



Key Technologies

HARDWARE	SOFTWARE	CONTROL
- Solar PV - MPPT - LiFePO₄ - BLDC/DC Fans - Compressor - ESP32 - Sensors	- Embedded C/C++ - IoT Dashboard - Data Logging	- Temperature-based automatic control - Battery monitoring - Alerts

Feasibility & Viability

FEASIBILITY

- System design is based on proven refrigeration principles
- All components are commercially available and affordable
- Peltier cooling is reliable for 20-40L capacity
- Solar + battery system is well-established technology
- ESP32 IoT integration is straightforward and scalable
- Estimated cost: ₹25,000 (affordable for small farmers)
- Build time: 2-3 weeks with standard tools

POTENTIAL CHALLENGES & RISKS

- Ambient temperature limits (Peltier less effective above 40°C)
- Battery degradation over 3-5 years
- Sensor calibration and maintenance
- User training for IoT dashboard
- Initial adoption resistance from traditional farmers
- Seasonal solar availability variations

STRATEGIES FOR OVERCOMING THEM

- Design thermal insulation to handle high ambient temps
- Use high-cycle LiFePO₄ batteries; plan replacement schedule
- Use robust, waterproof sensors; provide maintenance guides
- Create simple, intuitive dashboard; offer farmer training programs
- Demonstrate ROI through pilot projects and farmer testimonials
- Implement battery backup and hybrid charging options

Impact & Benefits

Our Smart Solar Cold Storage System offers multi-faceted benefits across social, economic, and environmental dimensions, transforming post-harvest practices for rural farmers.



SOCIAL IMPACT

- Accessible farm-level storage
- Reduced dependence on distant facilities
- Improved storage flexibility
- Potential rural employment opportunities



ECONOMIC IMPACT

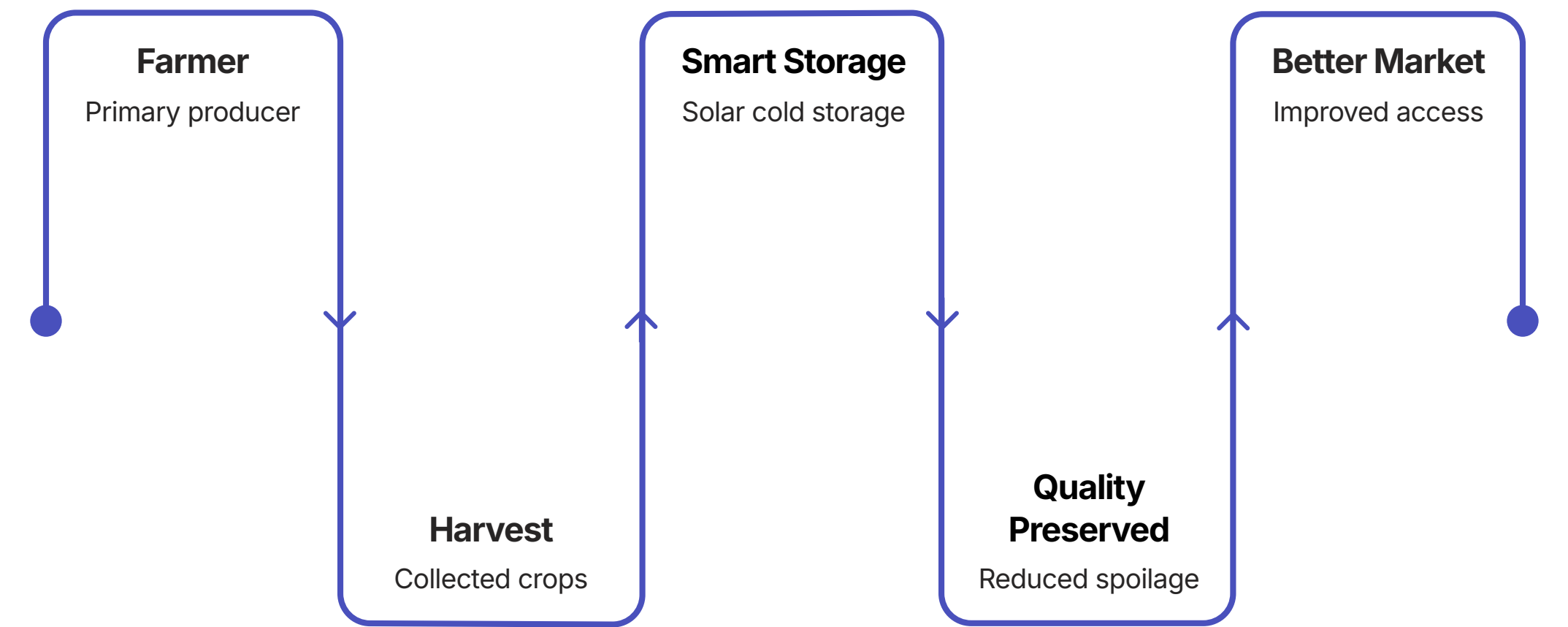
- Reduce avoidable post-harvest losses
- Preserve product quality
- Enable better timing of sales
- Increase value realization potential



ENVIRONMENTAL IMPACT

- Solar-powered operation
- Reduced grid dependency
- Efficient DC-powered architecture
- Reduced wastage of food resources

Enhanced Value Chain



Future Impact

Future Intelligence: IoT → Data Analytics → Shelf-Life Prediction → Demand-Aware Storage

📌 These are future enhancements, not currently implemented.

Research & References

RESEARCH FOUNDATION & EXISTING SOLUTIONS

- 1

PUSA SUNFRIDGE

 - Architecture: Solar-powered cold storage system
 - Solar Configuration: 50-100W PV panels
 - Storage Capacity: 50-100L
 - Cooling Mechanism: Vapor compression refrigeration
 - Intended Users: Small-scale farmers
 - Deployment Model: Decentralized farm-level storage
 - Our Addition: IoT monitoring, automatic control, optimized for 20-40L, Peltier-based cooling
- 2

COLD CHAIN & POST-HARVEST RESEARCH

 - Focus: Vegetable storage requirements
 - Temperature: 4-8°C for leafy greens, 10-15°C for root vegetables
 - Humidity: 85-95% for most vegetables
 - Post-harvest Deterioration: 30-40% losses without proper storage
- 3

NORTHEAST AGRICULTURAL CONTEXT

 - Focus: NER horticulture challenges
 - Cold-chain infrastructure gaps in rural areas
 - Farmer challenges: Limited access to storage, high spoilage rates
 - Post-harvest losses: 25-35% in NER region
- 4

REFRIGERATION RESEARCH

 - Focus: Solar refrigeration systems
 - DC/BLDC cooling efficiency
 - Thermal insulation optimization
 - Energy efficiency in off-grid systems
- 5

IoT & SMART AGRICULTURE

 - Focus: ESP32-based monitoring systems
 - Environmental monitoring in agriculture
 - IoT applications in cold storage
 - Smart cold storage solutions

BILL OF MATERIALS (BOM)

A detailed table with columns: Component | Specification | Quantity | Estimated Cost

Component	Specification	Quantity	Estimated Cost
Insulated box	20–40 L	1	₹2,000
Peltier cooling kit	12 V complete assembly	1	₹1,500
Extra 12 V fan	120 mm	1	₹350
Solar panel	100 W	1	₹3,500
Battery	12 V, 20–40 Ah	1	₹7,500
Charge controller	10 A	1	₹1,200
ESP32	Wi-Fi development board	2	₹900
DS18B20	Waterproof temperature sensor	2	₹400
SHT31/DHT22	Humidity sensor	1	₹450
Reed switch	Door sensor	1	₹100
INA219/INA226	Energy measurement	1	₹450
OLED	0.96-inch I²C	1	₹250
MOSFET/relay	Cooling control	1–2	₹300
Buck converter	12 V to 5 V	1	₹250
Buzzer, LEDs and buttons	Alerts/manual control	Set	₹300
Fuse, switch and distribution	Protection	Set	₹1,150
Wires and connectors	Cable, lugs, terminals, MC4	Set	₹1,600
Enclosure and mounting	PCB box, brackets, hardware	Set	₹1,200
Spares and contingency	Backup parts and delivery	—	₹4,000
Total	Complete system estimate	—	₹25,000